IEEE Access
Multidisciplinary : Rapid Review : Open Access Journal

Received 13 May 2025, accepted 5 June 2025, date of publication 30 June 2025, date of current version 10 July 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3584709

RESEARCH ARTICLE

Customer Lifetime Value Modeling via Two Stage Selected Trees Ensembles

ZARDAD KHAN[1], AMJAD ALI[1], SAEED ALDAHMANI[1], HENRIK NORDMARK[2,3], AND BERTHOLD LAUSEN[2,4]

[1]Department of Statistics and Business Analytics, United Arab Emirates University, Al Ain, United Arab Emirates
[2]School of Mathematics, Statistics and Actuarial Science (SMSAS), University of Essex, CO4 3SQ Colchester, U.K.
[3]Profusion Media Ltd., EC2A 4DP London, U.K.
[4]School of Medicine, Friedrich-Alexander University (FAU) of Erlangen-Nuremberg, 91054 Erlangen, Germany

Corresponding authors: Zardad Khan (zaar@uaeu.ac.ae) and Saeed Aldahmani (saldahmani@uaeu.ac.ae)

This work was supported by United Arab Emirates University, College of Business and Economics, the CBE Annual Research Program (CARP 2023). The work of Henrik Nordmark and Berthold Lausen was supported in part by the Business and Local Government Data Research Centre, funded by the Economic and Social Research Council, U.K., to provide researchers and analysts with secure data services, under Grant ES/L011859/1; in part by the Knowledge Transfer Partnership Program through Innovate U.K., University of Essex, under Grant Z699129X; and in part by Profusion Media Ltd. (a company registered in England), under Grant 6947442.

ABSTRACT The successful implementation of customer relationship management and product development need individual customer value assessment to generate sustained profit returns. As an important part of the customer relationship management, predicting future buyers and their monetary value allow companies to tailor interactions, focus on high-value prospects, and increase customer profitability and retention. This paper proposes a system that adopts two sequential stages to calculate customer lifetime value based on their purchase behaviors. To handle the diversity and imbalance in customer data, we explore three decision tree selection strategies to form an overall efficient ensemble. Decision tree selection is done based on individual tree performance on independent samples, out-of-bag samples, and sub-samples for their inclusion in the final ensemble. The first stage predicts whether a customer will purchase in a future window, while the second stage estimates the monetary value of their future purchases. Model evaluation includes Brier score, accuracy, kappa, and sensitivity for the first stage, and unexplained variation, RMSE, Spearman correlation, and MAD for the second stage. Based on a thorough analysis of five customer transaction datasets that are further divided into power and sporadic customers data, the proposed ensemble methods surpass several other state-of-the-art methods, i.e., k nearest neighbor, weighted kNN, random forests, support vector machine and neural networks in majority of the cases considering both the stages.

INDEX TERMS Customer lifetime value, random forest, churn prediction, monetary estimation.

I. INTRODUCTION

The strategic necessity for sustainable business development demands that organizations focus on understanding customer long-term value in today's competitive market. Customer Lifetime Value (CLTV) predicts the complete financial worth of a customer across their complete business association with a company through future forecasts. Organizations use CLTV instead of transactional metrics to track sustained revenue streams in that it gives them better information for making decisions about acquiring new customers, customer retention, and allocating resources.

Modern customer relationship management (CRM) systems in the present era have made CLTV estimation into an integral analytical process. High-value customers can be recognized early and their future value can be forecasted precisely by firms so they can optimize their marketing budgets and develop individualized promotional strategies and personalized experiences to build customer loyalty [1], [2], [3], [4], [5]. Company-driven insights from CLTV enable businesses to produce customer segmentations and create loyalty programs along with dynamic pricing approaches [6], [7], [8]. This predictive tool helps businesses reach maximum

The associate editor coordinating the review of this manuscript and approving it for publication was Ines Domingues.

© 2025 The Authors. This work is licensed under a Creative Commons Attribution 4.0 License.
For more information, see https://creativecommons.org/licenses/by/4.0/

profitability while helping them control customer loss rates by improving the management of their customer networks.

The estimation of CLTV requires applying data science methods to develop models that predict customer future behaviors based on transaction records and demographic information along with behavioral data. The prediction of customer future buying patterns including transaction volume and expense levels proves difficult because of fluctuating customer behaviors [5], [9], [10], [11]. The estimation of CLTV demands complex predictive modeling techniques that handle dual outcome types working for customer purchasing likelihood and monetary purchase value prediction.

Machine learning (ML) methods have been found to provide strong strategies to overcome the challenges faced by standard statistical models. Through the strategic use of ensemble learning, resampling techniques, and resilient classification-regression frameworks, modern ML approaches can effectively capture intricate, nonlinear patterns hidden within complex customers transactions data [12], [13].

This paper presents a CLTV system based on datasets from customers' transactions in various business scenarios using various features relating to individual customer purchase behaviors. The model consist of two stages. The first stage predicts customers who purchase in a future window and the second estimate a monetary value for the customers predicted to purchase in the first stage. Both the stages of the model are based on using selected decision trees enslembles. The proposed approach in the paper is novel in the following ways.

- The paper proposes using optimal random forest method to cater for the diverse nature of customer purchase behavior and dealing with class imbalanced problem in the data. Both these issue have an adverse effect on the overall performance of random forest due to non-informative classification and regression trees in the forest. Discarding trees with high error rates and integrating the rest of the optimal trees is thus intuitive to go for in such situation.
- The paper proposes the use of various descriptive measure, like standard deviation, skewness, kurtosis, etc., on the monetary values from each customers as predictors to capture the diverse patterns in customers' transaction data in conjunction with tree selection.
- The paper also evaluates feature selections by the suggested CLTV system in comparison with the standard random forest algorithm.

The rest of the paper is organized as follows. Section II gives a summary of the related work done in the literature. The proposed CLTV system is given in Section III. Section V gives a description of the datasets and the variables used to build the model. Experiments and results are given in Section VI. The paper ends with a conclusion given in Section VII.

II. RELATED WORK

Customer relationship management (CRM) has become the focus of research in academia and industry since the 1980s. It is due to the fact that maintaining a healthy relationship with individual customers can elevate profits and carries significant advantages in competing market [14], [15]. This has caused enterprises to undergo customer level competition [16], [17], [18]. This has also led researchers to understanding a customer's perceived value which can be defined as "the difference between total customer value and total customer cost" [7]. Jackson [19] initiated the concept of customer lifetime value (CLTV) utilizing the net present value method, that estimates the profits customers bring to businesses over their life cycle. Jackson's work can be considered as a groundwork for the research carried afterwards on the topic. Several researches have been dedicated to estimate a customer's lifetime value using models based on the recency, frequency and monetary (RFM) data [20], [21]. However, these models have several shortcomings. These models are most of the times impractical due to their conceptual nature and dependence on profit estimations. In the modern information era, where customers have various options and may not adhere to complete life cycles, the resulting data scarcity makes it difficult to calculate the profits and build effective models [22]. Consequently, due to the big data revolution, machine learning approaches have been observed to outperform other methods used for CLTV and churn prediction [23], [24], [25], [26], [27]. Examples of the various research endeavors include customers' churn prediction models such as classification trees [28], random forest [29], GAM [30] and support vector machine [31]. Chen and Fan [32] exploited support vector regression, customer behavior variables and controlled variables regarding multiple promotion for predicting dynamic customer lifetime value. Their study has found that machine learning models outperform the conventional interpretative models in terms of predictive capabilities.

Similarly, Ma and Xia [33] proposed a machine learning churn prediction model based on selected features for the telecom sector. They compared their model with logistic regression and decision tree and achieved better results. Using data from a similar domain, Vafeiadis et al. [34] conducted a comparative analysis of different machine learning methods tailored for customer churn prediction and showed support vector machine outperforming decision trees, artificial neural networks, and Bayesian classifiers using various performance metrics such as prediction hit rate, promotion coefficient and coverage. Martinez et al. [35] considered limit learning machine, logistic lasso regression, and gradient improvement trees for customers' future purchases prediction, showing that gradient improvement trees method has better prediction results than the others. The work of Jain et al. [36] has demonstrated the importance of feature extraction for improving the performance of machine learning algorithms. Further examples of related methods can be found in [37], [38], [39], and [40] and the references cited therein.

While methods proposed in the existing literature have demonstrated improved prediction performance, most of these methods are highly data dependent. Moreover, these methods overlook the impact of diverse data patterns that adversely affect the performance of machine learning models. Therefore, finding an optimal solution for churn prediction and monetary value estimation still needs further research. The current research is an attempt to develop a machine learning system based on tree selection strategies for the random forest that can effectively model customer transaction data capturing their diverse behavior. A brief comparison of the widely used machine learning methods and the proposed system is given in Table 1.

TABLE 1. Comparison of the standard models with the proposed method for CLV prediction.

Approach	Model Type	Handles Imbalance	Handling Diverse Patterns	Model Selection	Feature Importance	Limitation
k nearest neighbours (kNN)	Single Model	No	No	No	No	Sensitive to noise and irrelevant features [41]
Weighted kNN (WkNN)	Single Model	Partially	No	No	No	Weighting alone may not handle imbalance well [42], [43]
Random forest (RF)	Ensemble (Bagging)	Partially	No	No	Yes	No quality-based tree filtering [44], [45]
Support vector machine (SVM)	Single Model	No	No	No	No	Poor interpretability [46]
Neural network (NN)	Single Model	No	No	No	No	Requires large data; lacks transparency [47], [48]
Proposed optimal trees selection (OTE)	Ensemble (Pruned Trees)	Yes	Yes	Yes	Yes	High computational cost in ultra-large-scale data settings.

III. SELECTION STRATEGIES

A. OPTIMAL TREE SELECTION USING OUT-OF-BAG AND INDEPENDENT SAMPLES (OTE-OOB-IND)

Building on the methods presented in [49] and [50], let the training dataset be denoted as $\mathcal{D} = (X, Y)_{(d+1)\times n}$, where X is d dimensional matrix of feature space on n data points, and Y is binary response with labels 0 and 1. The OTE-OOB-IND method splits $\mathcal{D}$ into two disjoint subsets, $\mathcal{D}_{\mathcal{B}} = (X, Y)_B$, and $\mathcal{D}_{\mathcal{V}} = (X, Y)_V$ for trees fitting and its validation, respectively. This method follows the steps given below to create the optimized ensemble of trees:

1) A set of K trees are constructed using bootstrap samples drawn from $\mathcal{D}_{\mathcal{B}}$, using the Leo Breiman's random forest technique.
2) All the K trees constructed in Step 1 are sorted in ascending order based on their error rates computed on out-of-bag samples, and the most accurate top T trees are selected.
3) The T chosen trees are sequentially added, beginning with the top-ranked tree, while $\mathcal{D}_{\mathcal{V}}$ is utilized to assess the impact of each new tree on classification accuracy. Only those trees that enhance performance are kept, and the rest are discarded.
4) The retained trees are combined to form the final classification ensemble, which is used for prediction on unseen data.

While OTE has demonstrated improved classification accuracy over competing methods on several benchmark and simulated datasets [49], it can suffer from the loss of useful information. This issue arises because reserving a portion of the dataset for internal validation reduces the data available for model training. Previous research highlights that the performance of classification models is strongly linked to the amount of information utilized during training [51]. Two further pruning strategies are applied in this paper to make more effective use of the training data.

B. OPTIMAL TREE ASSESSMENT VIA OUT-OF-BAG OBSERVATIONS (OTE-OOB)

This approach utilizes out-of-bag data points for both individual and collective evaluation of trees. Empirical

evidence shows that, during bootstrap sampling, approximately one-third of the data is excluded from the sample [52]. These excluded observations can be repurposed as test instances for model evaluation. Let $\mathcal{B}_t$ represent the bootstrap sample, $\bar{\mathcal{B}}_t$ the corresponding out-of-bag sample, and $H(\mathcal{B}_t)$ the classification model trained on $\mathcal{B}_t$. Assume $\widehat{E}_t$ represents the error rate given by the $H(\mathcal{B}_t)$ on $\bar{\mathcal{B}}_t$. The final ensemble is created as follows:

1) Using the random forest technique, a total of K trees are constructed bootstrap samples, i.e, $\mathcal{B}_t$, where $t = 1, 2, \ldots, K$, and their respective error rates $\widehat{E}_t$ are determined.
2) The K trees generated in the previous step are ranked in ascending order based on their errors $\widehat{E}_t$, and the top T trees with the lowest errors are chosen. These selected trees are denoted as $H^{Q_1}(.), H^{Q_2}(.), \ldots, H^{Q_T}(.)$.
3) Beginning with $H^{Q_1}(.)$, each subsequent tree $H^{Q_j}(.)$, $j = 2, 3, \ldots, T$, is assessed individually using its associated out-of-bag sample. A tree is retained if:
$$\widehat{BrS}^{\langle j+\rangle} < \widehat{BrS}^{\langle j-\rangle}, \tag{1}$$
where, $\widehat{BrS}^{\langle j-\rangle}$ represents the Brier score given by the ensemble excluding the jth tree, while $\widehat{BrS}^{\langle j+\rangle}$ denotes the Brier score when the jth tree is included. The formula for calculating the Brier score is:
$$\widehat{BrS} = \frac{\sum_{i=1}^{n_v} \left(y_i - \widehat{P}(y_i = 1|X)\right)^2}{n_v}, \tag{2}$$
where $y_i \in \{0, 1\}$ represents the observed class label, n_v stands for the number of observations in the given validation data, and $\widehat{P}(y_i = 1|X)$ denotes the ensemble's probability prediction for the ith observation belonging to Class 1 given the features X.
In the context of regression, for monetory estimation in the second stage, let $\mathcal{V}_{exp}^{\langle j-\rangle}$ denote the explained variance of the ensemble without the jth tree, and let $\mathcal{V}_{exp}^{\langle j+\rangle}$ denote the explained variance of the ensemble including the kth tree. The tree $H^{Q_j}(.)$ is selected if the following condition holds:
$$\mathcal{V}_{exp}^{\langle j+\rangle} > \mathcal{V}_{exp}^{\langle j-\rangle}. \tag{3}$$
4) The selected trees are merged to build the final ensemble, which represents the optimized set of trees.

This process combines the strengths of the chosen trees, resulting in a robust and accurate ensemble model designed to provide superior performance on unseen data.

C. OPTIMAL TREE SELECTION VIA SUB-SAMPLES (OTE-SUB)

This approach involves selecting random sub-samples from the training dataset $\mathcal{D}$ without replacement for building trees [53]. The unused portion of each sub-sample is then employed to evaluate the performance of the corresponding classification tree. Define $\mathcal{S}_t$, $t = 1, 2, \ldots, K$, as sub-samples of size n' which is smaller than n, i.e., $n' < n$, where n represents the total number of instances in the training data, and let $\bar{\mathcal{S}}_t$ denote the remaining $n - n'$ observations. Let $H(\mathcal{S}_t)$ represent the tree constructed from $\mathcal{S}_t$, and $\widehat{E\text{sub}_t}$ its classification error when evaluated on $\bar{\mathcal{S}}_t$. The steps to create the final ensemble are as follows:

1) Instead of relying on bootstrap samples, K classification trees are generated using sub-samples $\mathcal{S}_t$, for $t = 1, 2, \ldots, K$. The error rates $\widehat{E\text{sub}_t}$ for each tree are evaluated.
2) The trees are arranged in ascending order based on their error rates $\widehat{E\text{sub}_t}$, and the best T trees are selected. These selected trees are denoted as $H^{Q_1}(.), H^{Q_2}(.), \ldots, H^{Q_T}(.)$.
3) Starting with the highest-ranked tree, each subsequent tree undergoes evaluation using its corresponding leftover sub-sample. Trees that satisfy the given Brier score/explained variance condition are retained, while the rest are excluded.
4) The final ensemble of optimized trees is constructed by merging the selected trees.

The sub-sample size n' should be carefully chosen to minimize tree correlation, as the number of possible combinations for building trees is limited to $\binom{n}{n'}$.

IV. THE TWO-STAGE SYSTEM

Exploiting the above selecting strategies, the stages of the proposed CLVT are described below.

STAGE 1: ESTIMATION OF PURCHASE PROPENSITY

The aim in this stage is to estimate the probability $\widehat{P}_i = \widehat{P}(y_i = 1 \mid \mathbf{X})$ of each customer making a purchase in a future window using the labeled dataset $\mathcal{D} = (X, Y)_{(d+1)\times n}$, where $\mathbf{Y} \in \{0, 1\}^n$ indicates whether a customer made a purchase, and $\mathbf{X}$ represents the feature matrix. To increase model robustness in the presence of diverse patterns in the customers' transaction data, the above mentioned three diverse and accurate classification tree selection strategies are used independently. The estimated binary class label $\widehat{y}_i = \mathbb{I}(\widehat{P}_i > \tau)$ is produced based on a predefine threshold τ, for example, $\tau = 0.5$.

STAGE 2: SPEND VALUE ESTIMATION

This stage predicts the expected monetary value $\widehat{V}_i = \mathbb{E}[\text{Spend}_i \mid \widehat{y}_i = 1]$ for those customers who are predicted to make a purchase in the given future window. The estimation is carried out using three separate regression strategies independently. The final output $\widehat{V}_i$ denotes the predicted monetary value for customer i.

An illustrative flowchart of the above two-stage system is given in Figure 1 and its pseudocode in Algorithm 1.

This paper examines the performance of alternative algorithms, i.e., k nearest neighbor (kNN) method [54], [55], its weighted variant (WkNN) [56], the neural network models (NN) [57], the random forest (RF) algorithm [58], and support vector machines (SVM) [59], against the proposed

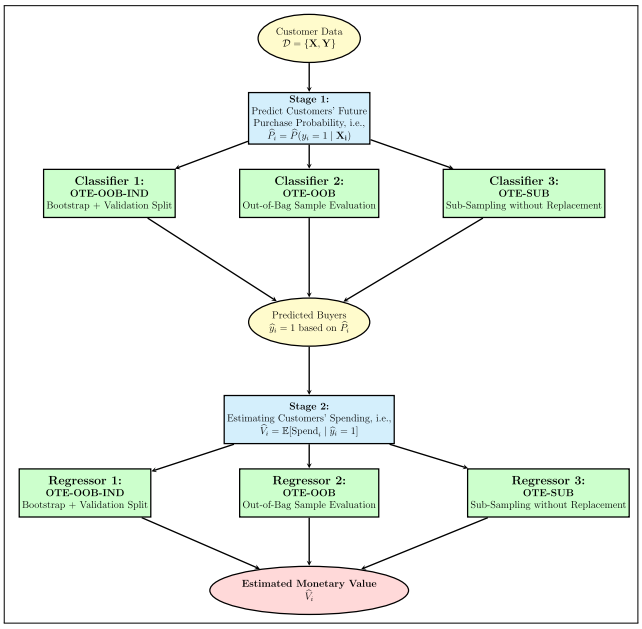

FIGURE 1. Overall flowchart of the proposed two-stage system based on the three tree selection strategies.

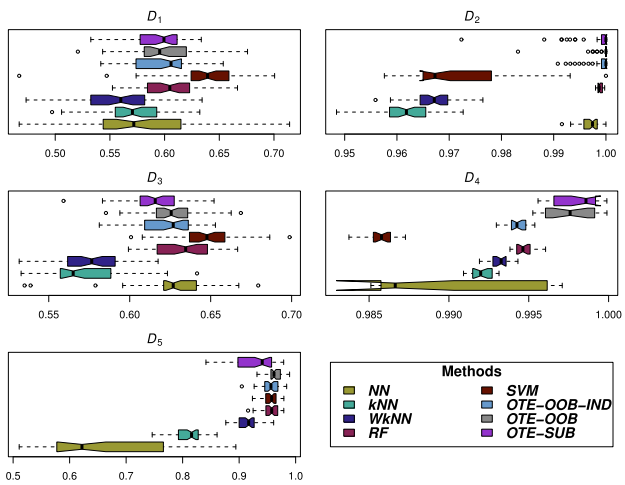

FIGURE 2. Classification accuracy of the methods on Power customers' data.

Algorithm 1 Two-Stage System Based on the Three Tree Selection Strategies for CLTV Prediction

Input: Customer data $\mathcal{D} = \{X, Y\}$, number of trees K, threshold τ
Output: Estimated customer value $\hat{V}_i$
Stage 1: Purchase Probability Estimation
for each strategy $\in$ {OTE-OOB-IND, OTE-OOB, OTE-SUB} **do**
  for $t = 1$ to K **do**
    Train classifier using the selected strategy:
    if strategy is OTE-OOB-IND **then**
      Use bootstrap sampling + validation split
    else if strategy is OTE-OOB **then**
      Use out-of-bag (OOB) evaluation
    else if strategy is OTE-SUB **then**
      Use sub-sampling without replacement
    end if
    Evaluate performance (OOB or validation)
  end for
  Rank classifiers and select top T models
  for each customer i **do**
    Compute $\widehat{P}_i = \widehat{P}(y_i = 1 \mid \mathbf{X})$
    Predict $\hat{y}_i = 1$ if $\widehat{P}_i \geq \tau$ else $\hat{y}_i = 0$
  end for
end for
Stage 2: Conditional Spending Estimation
Filter customers where $\hat{y}_i = 1$
for each strategy $\in$ {OTE-OOB-IND, OTE-OOB, OTE-SUB} **do**
  for $t = 1$ to K **do**
    Train regressor using selected strategy on $\{i : \hat{y}_i = 1\}$
    Evaluate regression performance (e.g., RMSE)
  end for
  Rank regressors and select top T models
  for each predicted buyer i **do**
    Compute $\widehat{V}_i = \mathbb{E}[\text{Spend}_i \mid \widehat{y}_i = 1]$
  end for
end for

strategies on customer transactions datasets given in the following section.

V. DESCRIPTION OF THE DATASETS AND THE FEATURES SET

This paper evaluates the performance of the proposed system against the others on a total of 5 big datasets. A description of the datasets is given in Table 2. Based on the number of transactions made by each customer, each of the 5 datasets are segmented into power and sporadic customers based on individual number of transactions above and below the median transactions, respectively, for all customers. This gave a total of 10 datasets. The main feature in each of these datasets is the number of days since the most recent purchase for a customer. This feature is usually called recency in conventional CLTV models. Other important features are the amount of monetary generated by a customer throughout his entire tenure and number of transactions made. Five other features are generated per customer from the set of all transactions' monetary. These are standard deviation, skewness, kurtosis, mean and trimmed mean of the monetary values of the transaction amounts by each customer. The reason for using these variables was to capture information on the diverse nature of customer purchase behaviour.

VI. EXPERIMENTS AND RESULTS

A. EXPERIMENTAL SETUP

Both the sporadic and power datasets are split following a chronological approach that entails arranging the data based on the time sequence, where earlier samples are

TABLE 2. Description of datasets.

ID	D ata	Source	Observations
D_1	cdnow	[60]	6696
D_2	Online retail dataset	https://www.kaggle.com/datasets/lakshmi25npathi/online-retail-dataset	33356
D_3	E-Commerce Data	https://www.kaggle.com/datasets/carrie1/ecommerce-data	20000
D_4	Bakery Sales Dataset	https://www.kaggle.com/datasets/akashdeepkuila/bakery	14460
D_5	Customer Segmentation	https://www.kaggle.com/code/fabiendaniel/customer-segmentation	240680

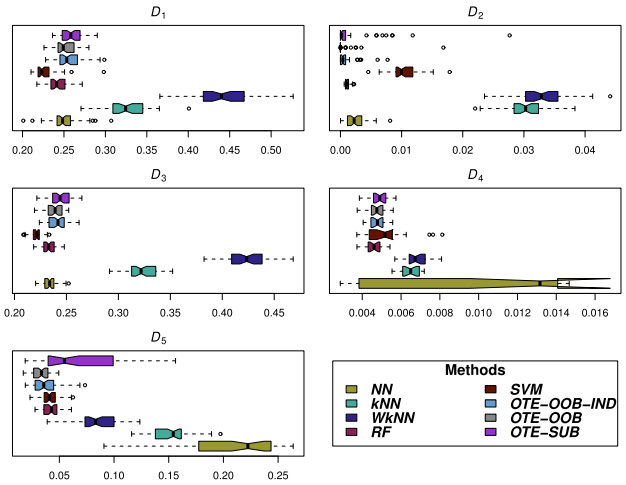

FIGURE 3. Brier score of the methods on Power customers' data.

utilized for training purposes and the later observations are used for testing purposes. This allows evaluating models' performance on more recent samples and simulates real world scenarios. The first 80% of observations, in the chronological order, are used to train the model while the last/recent 20% are used for performance assessment. For the three pruning strategies, a total of 1000 trees are grown in the initial ensemble and 50% of the top ranked trees are selected based on out-of-bag performance for the second stage model selection based on collective performance. Number of features taken at random at each node of the trees is set

TABLE 3. Classification results for Power customers who purchase in the given future window.

Metrics	Methods	Datasets				
		D_1	D_2	D_3	D_4	D_5
Accuracy	OTE-SUB	0.592	*0.998*	0.617	**0.998**	0.927
	OTE-OOB	*0.599*	**0.999**	0.627	**0.998**	**0.963**
	OTE-OOB-IND	0.597	**0.999**	0.622	0.994	*0.957*
	SVM	**0.636**	0.971	**0.650**	*0.996*	0.955
	RF	0.604	**0.999**	*0.633*	0.995	*0.957*
	WkNN	0.559	0.967	0.575	0.993	0.916
	kNN	0.570	0.962	0.572	0.992	0.812
	NN	0.583	0.997	0.627	0.990	0.668
Kappa	OTE-SUB	0.168	0.973	0.150	*0.999*	0.850
	OTE-OOB	0.179	**0.990**	*0.167*	**1.000**	**0.925**
	OTE-OOB-IND	0.177	0.982	0.160	0.977	0.911
	SVM	**0.249**	0.197	0.123	0.058	0.908
	RF	*0.191*	*0.986*	**0.172**	0.991	*0.911*
	WkNN	0.103	0.419	0.091	0.058	0.827
	kNN	0.128	0.275	0.074	0.081	0.613
	NN	0.117	0.954	0.053	0.029	0.268
Sensitivity	OTE-SUB	0.681	0.980	0.388	*0.998*	0.935
	OTE-OOB	0.700	**0.991**	0.387	**1.000**	0.950
	OTE-OOB-IND	0.692	0.989	*0.393*	0.993	**0.954**
	SVM	**0.780**	0.159	0.160	0.038	0.952
	RF	0.703	*0.990*	0.368	0.989	0.951
	WkNN	0.624	0.401	**0.421**	0.059	0.937
	kNN	0.633	0.249	*0.393*	0.070	0.853
	NN	*0.777*	0.964	0.106	0.044	0.867
BS	OTE-SUB	0.250	*0.002*	0.245	**0.000**	0.069
	OTE-OOB	0.253	**0.001**	0.239	**0.000**	**0.033**
	OTE-OOB-IND	0.256	**0.001**	0.241	**0.000**	*0.038*
	SVM	**0.228**	0.010	**0.221**	*0.003*	0.041
	RF	*0.244*	0.002	*0.233*	0.001	0.043
	WkNN	0.441	0.033	0.425	0.006	0.084
	kNN	0.326	0.030	0.323	0.004	0.152
	NN	0.251	0.003	0.234	*0.003*	0.207

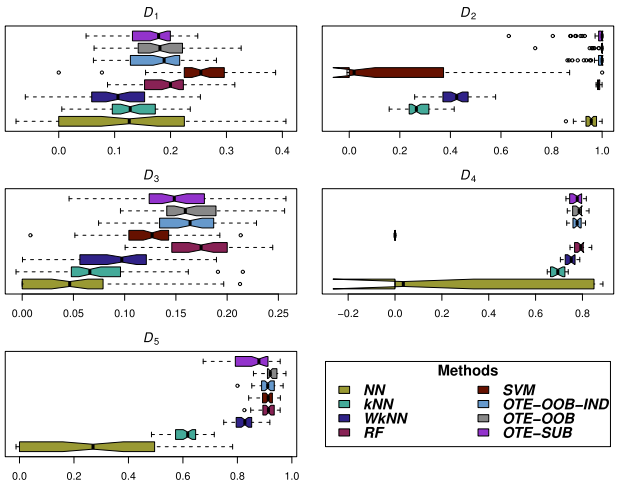

FIGURE 4. Kappa values of the methods on Power customers' data.

to the square root of the total number of features. Trees are grown unpruned for Stage 1, and at Stage 2, node size is set to a minimum of 5 observations in the leaf nodes. For SVM, we used the `ksvm` function in the `kernlab` [61]

TABLE 4. Classification results for Sporadic customers who purchase in the given future window.

Metrics	Methods	Datasets				
		D_1	D_2	D_3	D_4	D_5
Accuracy	OTE-SUB	0.8342	0.9994	0.8045	**0.9999**	**0.9798**
	OTE-OOB	0.8358	**0.9999**	0.8156	0.9938	0.9727
	OTE-OOB-IND	0.8315	0.9972	0.7979	0.9998	0.9659
	SVM	**0.8530**	0.9681	0.8250	0.9895	0.8133
	RF	0.8444	0.9984	0.8258	0.9997	0.9746
	WkNN	0.7662	0.9848	0.7220	0.9965	0.9627
	kNN	0.7949	0.9649	0.7476	0.9942	0.8310
	NN	0.8519	0.9917	0.8280	0.9986	0.7705
Kappa	OTE-SUB	**0.0887**	0.9894	0.0759	0.9928	**0.9497**
	OTE-OOB	0.0788	**0.9986**	0.0818	**0.9955**	0.9359
	OTE-OOB-IND	0.0882	0.9464	0.0748	0.9834	0.9187
	SVM	0.0005	0.0394	0.0002	0.9902	0.6251
	RF	0.0673	0.9670	0.0882	0.9715	0.9395
	WkNN	0.0625	0.6960	0.0368	0.6951	0.9128
	kNN	0.1068	0.1873	0.0290	0.4860	0.5966
	NN	0.0543	0.8095	0.0415	0.7824	0.2860
Sensitivity	OTE-SUB	0.1035	0.9847	0.1032	0.9870	0.9545
	OTE-OOB	0.0931	**0.9975**	0.0878	**0.9956**	**0.9570**
	OTE-OOB-IND	0.1105	0.9545	0.1146	0.9929	0.9469
	SVM	0.0003	0.0432	**0.0000**	0.9906	0.7957
	RF	0.0671	0.9523	0.0739	0.9486	0.9462
	WkNN	0.1998	0.6439	0.2058	0.6850	0.9547
	kNN	**0.2010**	0.1606	0.1524	0.4756	0.7040
	NN	0.0445	0.8307	0.0301	0.7842	0.3480
BS	OTE-SUB	0.1335	0.0017	0.1670	**0.0001**	**0.0217**
	OTE-OOB	0.1299	**0.0003**	0.1613	**0.0001**	0.0235
	OTE-OOB-IND	0.1342	0.0020	0.1681	0.0002	0.0295
	SVM	0.1252	0.0209	0.1448	0.0046	0.1538
	RF	0.1264	0.0045	0.1562	0.0005	0.0458
	WkNN	0.2338	0.0152	0.2780	0.0035	0.0373
	kNN	0.1671	0.0294	0.2063	0.0043	0.1199
	NN	**0.1241**	0.0078	**0.1434**	0.0014	0.1675

TABLE 5. Results for power customers monetary prediction.

Metrics	Methods	Datasets				
		D_1	D_2	D_3	D_4	D_5
Spearman correlation	OTE-SUB	0.957	0.726	0.974	**0.988**	0.898
	OTE-OOB	*0.958*	*0.793*	*0.975*	*0.987*	*0.923*
	OTE-OOB-IND	**0.963**	**0.838**	*0.975*	*0.987*	**0.933**
	SVM	0.881	0.538	0.825	0.986	0.653
	RF	0.953	0.786	**0.985**	0.987	0.893
	WkNN	0.758	0.694	0.837	0.974	0.695
	kNN	0.620	0.817	0.772	0.983	0.883
	NN	0.370	0.445	0.350	0.228	0.182
Unexplained variation	OTE-SUB	0.268	0.454	0.102	*0.051*	0.372
	OTE-OOB	*0.254*	0.305	*0.087*	0.055	*0.311*
	OTE-OOB-IND	**0.245**	**0.249**	**0.082**	**0.050**	**0.284**
	SVM	0.620	0.928	0.762	0.149	0.835
	RF	0.299	0.308	0.105	0.056	0.365
	WkNN	0.378	*0.276*	0.178	0.089	0.329
	kNN	0.646	0.293	0.236	0.070	0.338
	NN	0.882	0.962	0.773	0.954	0.947
MAD	OTE-SUB	*0.020*	0.045	**0.003**	**0.012**	0.015
	OTE-OOB	*0.020*	*0.037*	**0.003**	**0.012**	*0.014*
	OTE-OOB-IND	**0.018**	**0.034**	**0.003**	**0.012**	**0.013**
	SVM	0.031	0.134	0.010	*0.013*	0.026
	RF	0.025	0.041	*0.004*	0.013	0.015
	WkNN	0.035	0.038	0.006	0.017	0.018
	kNN	0.046	0.038	0.008	0.014	*0.014*
	NN	0.059	0.091	0.017	0.070	0.039
RMSE	OTE-SUB	0.051	0.086	*0.017*	**0.022**	0.056
	OTE-OOB	*0.050*	0.072	**0.016**	*0.023*	*0.052*
	OTE-OOB-IND	**0.048**	**0.046**	**0.016**	**0.022**	**0.050**
	SVM	0.080	0.172	0.049	0.038	0.086
	RF	0.055	0.078	0.018	*0.023*	0.057
	WkNN	0.060	*0.064*	0.023	0.029	0.052
	kNN	0.079	0.080	0.027	0.026	0.054
	NN	0.100	0.172	0.049	0.096	0.095

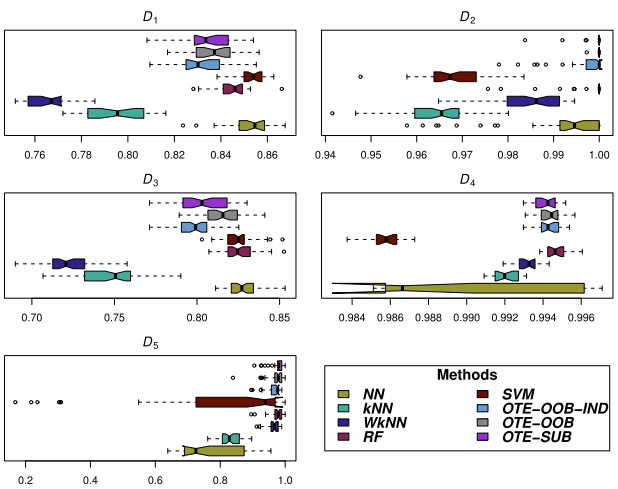

FIGURE 5. Classification accuracy of the methods on Sporadic customers' data.

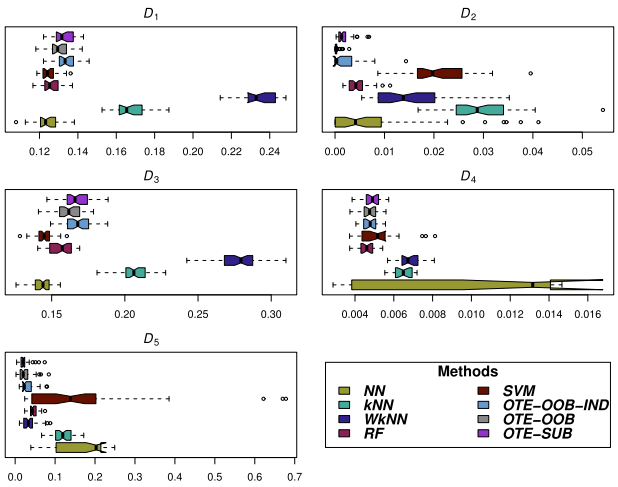

FIGURE 6. Brier score of the methods on Sporadic customers' data.

R package using a linear kernel with default settings for the rest of the parameters. For random forest, we used the implementation of Liaw and Wiener [62], `randomForest`, in the R programming language. For neural network and weighted kNN, the R packages `nnet` [63] and `kknn` [64] are used, respectively. Results for power and sporadic customers are given in Tables 3 and 4, respectively. The tables give Stage 1 and Stage 2 assessment of the methods on power and

TABLE 6. Results for sporadic customers monetary prediction.

Metrics	Methods	Datasets				
		D_1	D_2	D_3	D_4	D_5
Spearman correlation	OTE-SUB	*0.987*	0.812	0.833	*0.996*	0.978
	OTE-OOB	*0.987*	0.821	0.834	*0.996*	0.978
	OTE-OOB-IND	**0.988**	*0.822*	*0.836*	0.995	**0.989**
	SVM	0.968	0.682	0.666	0.985	0.577
	RF	0.986	0.764	**0.886**	0.995	0.892
	WkNN	0.946	0.462	0.502	0.995	0.751
	kNN	0.877	**0.864**	0.830	**0.997**	0.970
	NN	0.643	0.354	0.414	0.437	0.139
Unexplained variation	OTE-SUB	0.133	0.192	0.278	0.017	*0.497*
	OTE-OOB	*0.123*	*0.169*	0.262	0.015	0.504
	OTE-OOB-IND	**0.111**	**0.148**	**0.242**	**0.013**	**0.484**
	SVM	0.252	0.932	0.624	0.205	0.937
	RF	0.149	0.346	0.280	*0.014*	0.588
	WkNN	0.148	0.330	*0.252*	0.015	0.540
	kNN	0.327	0.176	0.255	0.017	0.518
	NN	0.756	0.946	0.882	0.796	0.846
MAD	OTE-SUB	*0.011*	0.028	*0.010*	**0.001**	**0.013**
	OTE-OOB	*0.011*	*0.026*	**0.009**	**0.001**	**0.013**
	OTE-OOB-IND	**0.010**	**0.024**	*0.010*	**0.001**	**0.013**
	SVM	0.016	0.134	0.015	0.007	0.026
	RF	0.014	0.037	0.011	0.010	0.017
	WkNN	0.019	0.039	0.013	*0.002*	0.017
	kNN	0.032	0.027	0.010	*0.002*	0.014
	NN	0.038	0.095	0.031	0.037	*0.037*
RMSE	OTE-SUB	0.032	0.069	0.039	*0.009*	*0.068*
	OTE-OOB	*0.031*	*0.063*	*0.038*	**0.008**	*0.068*
	OTE-OOB-IND	**0.029**	**0.058**	**0.037**	**0.008**	**0.067**
	SVM	0.045	0.172	0.060	0.032	0.089
	RF	0.035	0.084	0.040	0.009	0.075
	WkNN	0.035	0.088	*0.038*	0.010	0.071
	kNN	0.052	0.070	*0.038*	*0.009*	0.071
	NN	0.065	0.170	0.083	0.062	0.097

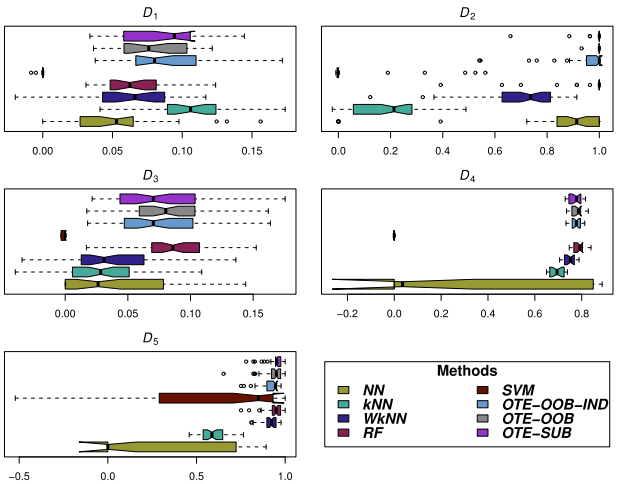

FIGURE 7. Kappa values of the methods on Sporadic customers' data.

sporadic customers data. Brier score, classification accuracy, sensitivity and Cohen's kappa are estimated for Stage 1 where as Stage 2 performance is assessed via unexplained variations, RMSE, Spearman correlation coefficient and MAD.

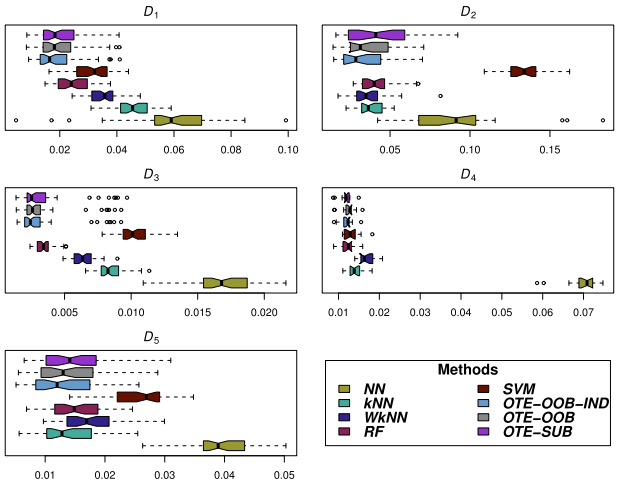

FIGURE 8. MAD of the methods on Power customers' data.

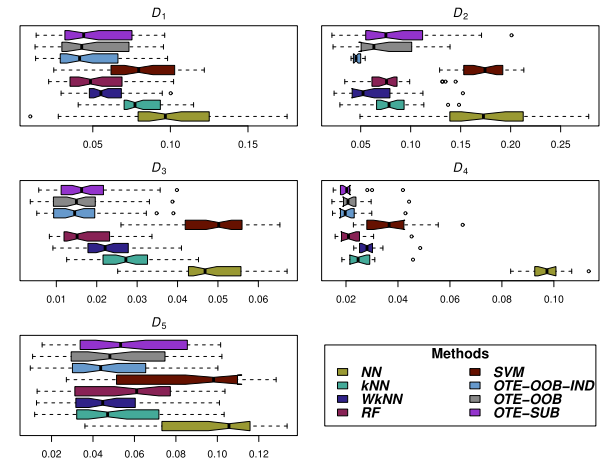

FIGURE 9. RMSE of the methods on Power customers' data.

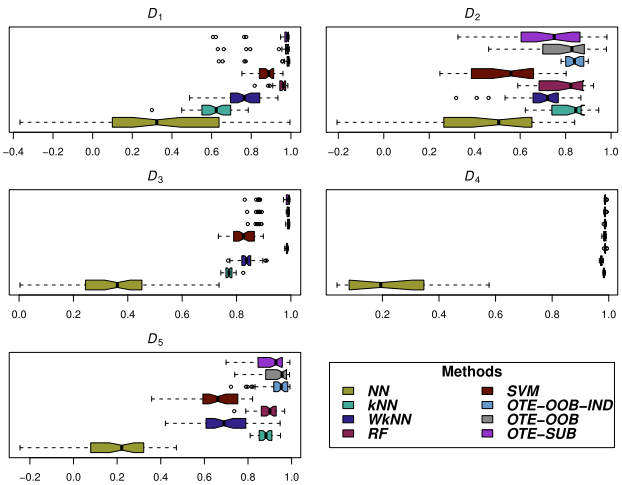

FIGURE 10. Spearman correlation of the methods on Power customers' data.

B. RESULTS

Tables 3 and 4 show results on Power and Sporadic customers' data, respectively, based on predicting customers who purchase in the given future window. The tables list

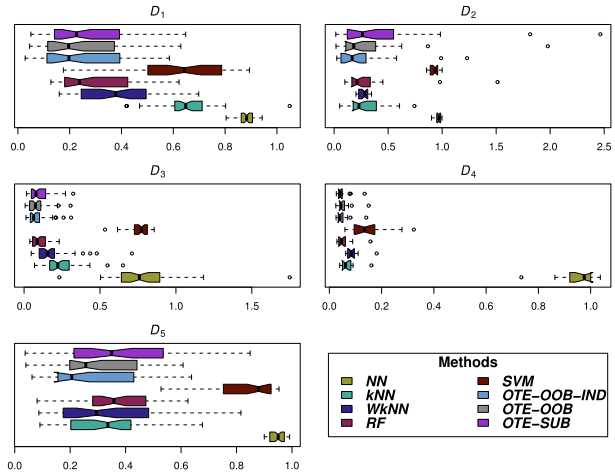

FIGURE 11. Unexplained variation of the methods on Power customers' data.

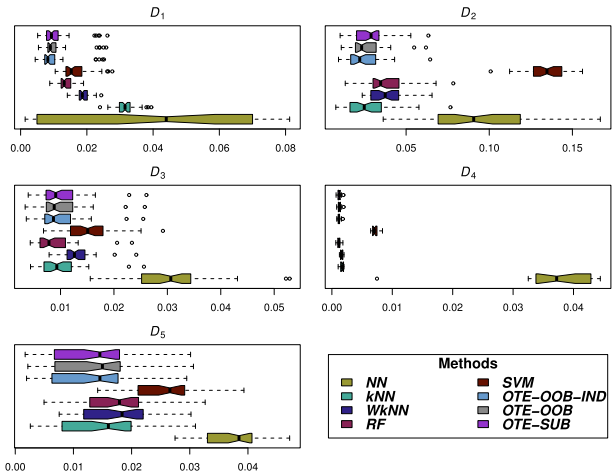

FIGURE 12. MAD of the methods on Sporadic customers' data.

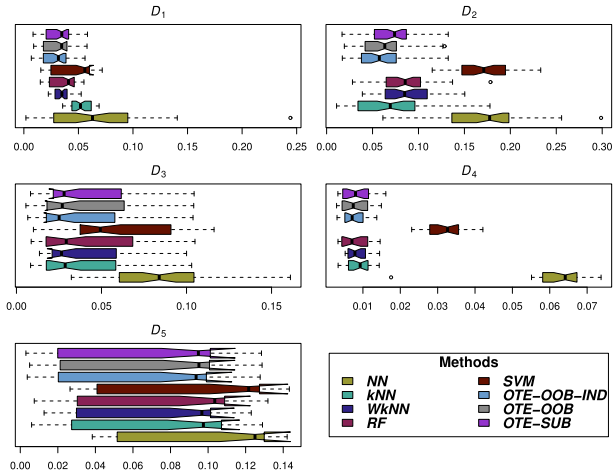

FIGURE 13. RMSE of the methods on Sporadic customers' data.

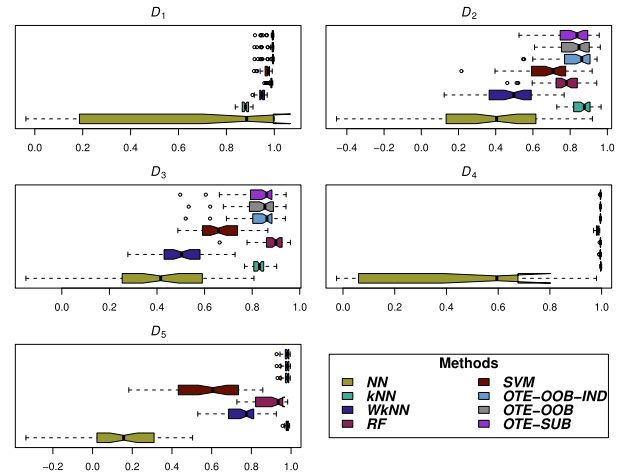

FIGURE 14. Spearman correlation of the methods on Sporadic customers' data.

classification accuracy, Cohen's kappa, sensitivity and Brier score values against the various methods considered on the datasets. Result of the best performing method is shown in bold. Table 3 shows that the proposed CLTV system demonstrates better results than the other on 3 out of the 5 power customrers' datasets considering all the performance metrics. None of the other methods has outperformed on any of the datasets. Similarly, based on the result given in Table 4, the suggested system demonstrates better results than the other on 3 out of 5 Sporadic customers' datasets based on all the performance metrics. None of the other methods have won consistently, considering all the performance metrics, on any of the datasets.

For monetary value estimation assessment of customer who purchase in the given future window, the result are given in Tables 5 and 6, for Power and Sporadic customers, respectively. The tables give results for each of the 5 datasets (Power and Sporadic), in terms of Spearman correlation, Unexplained variation, MAD and RMSE. In the case of Power customers' datasets, it is evident from Table 5 that the proposed system outperforms the other methods consistently in terms of unexplained variation, MAD and RMSE. In terms of Spearmna correlation, the CLTV system outperforms the other method on 4 out of 5 datasets, with random forest being the best performing method on one dataset. In the case of Sporadic datasets, it is evident from Table 6 that the proposed method consistently outperforms the other methods on 4 out of 5 datasets in terms of unexplained variation, MAD and RMSE. In terms of Spearman correlation, the proposed system is giving better results than the others on 2 out of 5 datasets, whereas kNN and random forest gave better results than the others on 2 and 1 datasets, respectively.

For additional insights into the results, boxplots of the results are shown in Figures 2-14. These results are obtained from 500 runs based on random 80% and 20% training and testing partitioning, respectively, of the given data unlike the results whown in the tables where historic 80% observations were used for training while the more recent 20% observations for testing. From these figures, similar conclusions, as given above, can be drawn. This demonstrates that proposed CLTV system outperforms the rest of the methods in most of the cases.

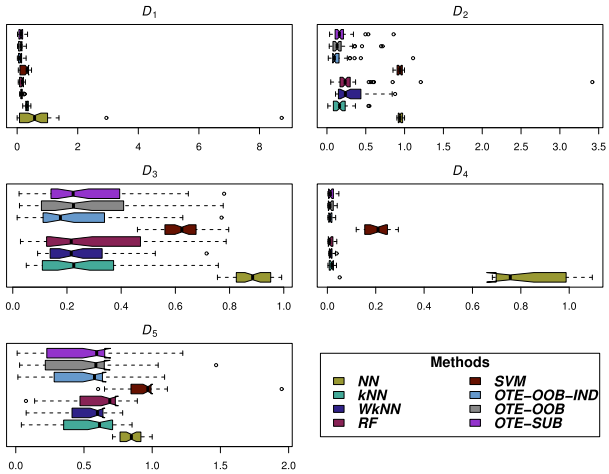

FIGURE 15. Unexplained variation of the methods on Sporadic customers' data.

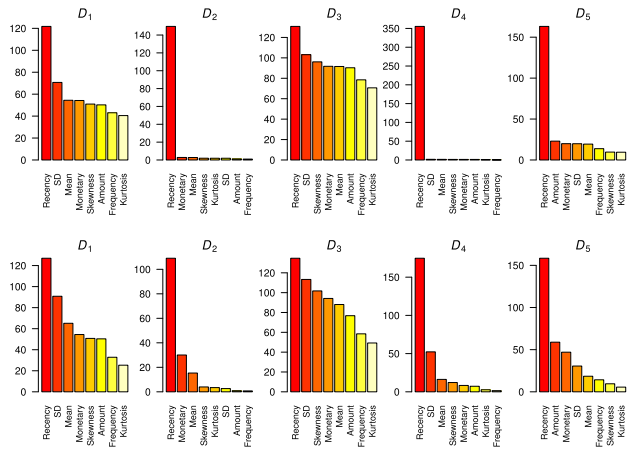

FIGURE 16. Barplots for variable importance in terms of mean Gini decrease. The top panel shown random forest variable importance and the bottom panel shows variable importance by the proposed CLTV system. Recency can be observed to be the most important variable by both the methods. The proposed system allows other variable to take part in model construction.

The proposed CLTV systems has also been used for assessing informative features in comparison with random forest. Mean decrease in the Gini index has been used as a feature importance measure. Figure 16 shows the feature importance plot for random forest and the proposed CLTV system (OTE-OOB-SUB). The top panel shows feature impotence by random forest whereas the bottom panel gives feature importance by the proposed system. It is evident from the plot that random forest uses mainly Recency in the ensemble as the most important feature while giving little significance to the rest of the features in model construction. On the other hand, the proposed system has considered other features, such as mean amount, standard deviation and monetary values from each customer transactions, as important by allowing them to take part in model construction. This demonstrates that the proposed model can effectively learn customers transactions data incorporating their diverse behaviours.

VII. CONCLUSION

A two stage CLTV system is proposed based on the idea of pruning the ensemble of classification and regression trees using three different strategies. The system is compared with neural networks, kNN, weighted kNN, SVM and random forest. The proposed method outperforms the other methods on most of the datasets.

The analysis given in the paper have demonstrated that selecting the best performing trees based on their individual and collective performance can effectively be used in churn prediction and monetary value estimation. This is due to the fact that customers' transaction data usually have diverse patterns making it hard for conventional methods to capture the underlying patterns effectively. The proposed system, as it is based on trees selection, discards trees that are adversely affected by data inconsistencies, as is the case with customers' transaction data. Based on the analysis given, it can therefore be concluded that the proposed CLTV system can give promising results both in churn prediction and monetary estimation while dealing with transactions data having diverse patterns.

In future we aim to consider different machine learning methods in Stage 1 & 2. The idea will be to first find the best machine learning method out of the available options at individual stages and then use the resultant methods at their corresponding stages to get an overall better model. This will be entirely data driven approach and businesses and firms might come up with different selections at individual stages for their datasets. Examples of possible machine learning methods for consideration in this regard could be as those given in [65], [66], [67], and [68]. Feature selection and the use of various socio-economic, engagement based, and demographic features could also be considered to see how it affects the performance of the models given in the paper. For additional randomness in the base models, the idea of randomly projecting the given feature space for each base model could also be incorporated in the proposed system.

REFERENCES

[1] Y. Yan and N. Resnick, "A high-performance turnkey system for customer lifetime value prediction in retail brands: Forthcoming in quantitative marketing and economics," *Quant. Marketing Econ.*, vol. 22, pp. 169–192, Nov. 2023.

[2] Y. S. Cho, N. Edevbie, A. K. W. Lau, and P. C. Hong, "The impact of social media technologies on supplier and customer relationship management: An empirical analysis in the us," *IEEE Access*, vol. 8, pp. 220878–220888, 2020.

[3] S. Wu, W.-C. Yau, T.-S. Ong, and S.-C. Chong, "Integrated churn prediction and customer segmentation framework for Telco business," *IEEE Access*, vol. 9, pp. 62118–62136, 2021.

[4] R. H. Khan, D. F. Dofadar, M. G. R. Alam, M. Siraj, M. R. Hassan, and M. M. Hassan, "LRFS: Online Shoppers' behavior-based efficient customer segmentation model," *IEEE Access*, vol. 12, pp. 96462–96480, 2024.

[5] K. Kim, M. Jo, I. Ra, and S. Park, "RFMVDA: An enhanced deep learning approach for customer behavior classification in e-commerce environments," *IEEE Access*, vol. 13, pp. 12527–12541, 2025.

[6] V. A. Zeithaml, K. N. Lemon, and R. T. Rust, *Driving Customer Equity: How Customer Lifetime Value is Reshaping Corporate Strategy*. New York, NY, USA: Simon and Schuster, 2001.

[7] H. Zhang, X. Liang, and S. Wang, "Customer value anticipation, product innovativeness, and customer lifetime value: The moderating role of advertising strategy," *J. Bus. Res.*, vol. 69, no. 9, pp. 3725–3730, Sep. 2016.

[8] H.-H. Zhao, X.-C. Luo, R. Ma, and X. Lu, "An extended regularized K-Means clustering approach for high-dimensional customer segmentation with correlated variables," *IEEE Access*, vol. 9, pp. 48405–48412, 2021.

[9] M. S. Kasem, M. A. Hamada, and I. Taj-Eddin, "Customer profiling, segmentation, and sales prediction using AI in direct marketing," *Neural Comput. Appl.*, vol. 36, no. 9, pp. 4995–5005, Dec. 2023.

[10] A. O. Dandis, M. Al Haj Eid, D. Griffin, R. Robin, and A. K. Ni, "Customer lifetime value: The effect of relational benefits, brand experiences, quality, satisfaction, trust and commitment in the fast-food restaurants," *TQM J.*, vol. 35, no. 8, pp. 2526–2546, Dec. 2023.

[11] S. Zhao, R. Wu, J. Tao, M. Qu, M. Zhao, C. Fan, and H. Zhao, "PerCLTV: A general system for personalized customer lifetime value prediction in online games," *ACM Trans. Inf. Syst.*, vol. 41, no. 1, pp. 1–29, Apr. 2022.

[12] R. Roy, P. Banerjee, S. K. Das, N. Dana, S. Roy, M. A. Amin, and M. Hasan, "Customer lifetime value prediction using machine learning techniques and its impact on FinTech," in *Advances in Science, Engineering and Technology*. Oxon, U.K: CRC Press, Apr. 2025, pp. 248–253.

[13] J. Akter, A. Roy, S. Rahman, S. Mohona, and J. Ara, "Artificial intelligence-driven customer lifetime value (CLV) forecasting: Integrating rfm analysis with machine learning for strategic customer retention," *J. Comput. Sci. Technol. Stud.*, vol. 7, no. 1, pp. 249–257, 2025.

[14] M. Mohan, M. W. Nyadzayo, and R. Casidy, "Customer identification: The missing link between relationship quality and supplier performance," *Ind. Marketing Manage.*, vol. 97, pp. 220–232, Aug. 2021.

[15] K. Gangadharan, A. Purandaran, M. Kanagasabai, B. Subramanian, R. Jeyaraj, and S. K. Jung, "From data to decisions: The power of machine learning in business Recommendations," *IEEE Access*, vol. 13, pp. 17354–17397, 2025.

[16] K. K. H. Kunasekaran, Y. Zheng, and W. Wang, "Research on customer relationship management based on data mining," *Asia–Pacific J. Convergent Res. Interchange*, vol. 6, no. 5, pp. 61–77, May 2020.

[17] Y.-W. Huang and H.-T. Tsou, "Empirical study of the affecting statistical education on customer relationship management and customer value in hi-tech industry," *EURASIA J. Math., Sci. Technol. Educ.*, vol. 14, no. 4, pp. 1287–1294, Jan. 2018.

[18] Q. Zhao, G. Mao, and S. Wang, "Balancing profit and cultural heritage: Multi-objective dynamic pricing for hanfu using deep deterministic policy gradient," *IEEE Access*, vol. 13, pp. 1–15, 2025.

[19] J. B. Bund, "Build customer relationship that last," *Harvard Bus. Rev.*, vol. 63, no. 6, pp. 28–120, 1985.

[20] L. Paul and T. R. Ramanan, "An RFM and CLV analysis for customer retention and customer relationship management of a logistics firm," *Int. J. Appl. Manage. Sci.*, vol. 11, no. 4, p. 333, Jan. 2019.

[21] H. Abbasimehr and M. Shabani, "A new methodology for customer behavior analysis using time series clustering: A case study on a bank's customers," *Kybernetes*, vol. 50, no. 2, pp. 221–242, 2021.

[22] Y. Sun, H. Liu, and Y. Gao, "Research on customer lifetime value based on machine learning algorithms and customer relationship management analysis model," *Heliyon*, vol. 9, no. 2, Feb. 2023, Art. no. e13384.

[23] A. Vanderveld, A. Pandey, A. Han, and R. Parekh, "An engagement-based customer lifetime value system for e-commerce," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, Aug. 2016, pp. 293–302.

[24] M. Afzal, S. Rahman, D. Singh, and A. Imran, "Cross-sector application of machine learning in telecommunications: Enhancing customer retention through comparative analysis of ensemble methods," *IEEE Access*, vol. 12, pp. 115256–115267, 2024.

[25] S. Saha, C. Saha, M. M. Haque, M. G. R. Alam, and A. Talukder, "ChurnNet: Deep learning enhanced customer churn prediction in telecommunication industry," *IEEE Access*, vol. 12, pp. 4471–4484, 2024.

[26] A. Manzoor, M. A. Qureshi, E. Kidney, and L. Longo, "A review on machine learning methods for customer churn prediction and recommendations for business practitioners," *IEEE Access*, vol. 12, pp. 70434–70463, 2024.

[27] K. Dhanushkodi, A. Bala, N. Kodipyaka, and V. Shreyas, "Customer behavior analysis and predictive modeling in supermarket retail: A comprehensive data mining approach," *IEEE Access*, vol. 13, pp. 2945–2957, 2025.

[28] J. Vijaya and E. Sivasankar, "Computing efficient features using rough set theory combined with ensemble classification techniques to improve the customer churn prediction in telecommunication sector," *Computing*, vol. 100, no. 8, pp. 839–860, Jun. 2018.

[29] J. Zhang, M. Zhi, and Y. Zhang, "Combined generalized additive model and random forest to evaluate the influence of environmental factors on phytoplankton biomass in a large eutrophic lake," *Ecological Indicators*, vol. 130, Nov. 2021, Art. no. 108082.

[30] K. Coussement, D. F. Benoit, and D. V. d. Poel, "Preventing customers from running away! Exploring generalized additive models for customer churn prediction," in *Proc. Acad. Marketing Sci. (AMS) Annu. Conf.*, Oct. 2014, p. 238.

[31] D. Boughaci and A. A.-S. Alkhawaldeh, "Three local search-based methods for feature selection in credit scoring," *Vietnam J. Comput. Sci.*, vol. 5, no. 2, pp. 107–121, May 2018.

[32] Z.-Y. Chen and Z.-P. Fan, "Dynamic customer lifetime value prediction using longitudinal data: An improved multiple kernel SVR approach," *Knowl.-Based Syst.*, vol. 43, pp. 123–134, May 2013.

[33] W. Ma and G. Xia, "Prediction model of customer churn based on deep neural network," *Comput. Technol. Dev*, vol. 29, pp. 76–80, 2019.

[34] T. Vafeiadis, K. Diamantaras, G. Sarigiannidis, and K. C. Chatzisavvas, "A comparison of machine learning techniques for customer churn prediction," *Simul. Model. Pract. Theory*, vol. 55, pp. 1–9, Apr. 2015.

[35] A. Martínez, C. Schmuck, S. Pereverzyev, C. Pirker, and M. Haltmeier, "A machine learning framework for customer purchase prediction in the non-contractual setting," *Eur. J. Oper. Res.*, vol. 281, no. 3, pp. 588–596, May 2018.

[36] H. Jain, A. Khunteta, and S. Srivastava, "Telecom churn prediction and used techniques, datasets and performance measures: A review," *Telecommun. Syst.*, vol. 76, no. 4, pp. 613–630, Oct. 2020.

[37] M. S. Kılınç and R. Rohrhirsch, "Predicting customers' cross-buying decisions: A two-stage machine learning approach," *J. Bus. Analytics*, vol. 6, no. 3, pp. 180–187, Sep. 2022.

[38] C. Tsai, Y. Hu, C. Hung, and Y. Hsu, "A comparative study of hybrid machine learning techniques for customer lifetime value prediction," *Kybernetes*, vol. 42, no. 3, pp. 357–370, Mar. 2013.

[39] A. Griva, E. Zampou, V. Stavrou, D. Papakiriakopoulos, and G. Doukidis, "A two-stage business analytics approach to perform behavioural and geographic customer segmentation using e-commerce delivery data," *J. Decis. Syst.*, vol. 33, no. 1, pp. 1–29, Jan. 2024.

[40] S. A. Curiskis, X. Dong, F. Jiang, and M. Scarr, "A novel approach to predicting customer lifetime value in B2B SaaS companies," *J. Marketing Analytics*, vol. 11, no. 4, pp. 587–601, Jul. 2023.

[41] S. Garcia, J. Derrac, J. Cano, and F. Herrera, "Prototype selection for nearest neighbor classification: Taxonomy and empirical study," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 34, no. 3, pp. 417–435, Mar. 2012.

[42] H. He and E. A. Garcia, "Learning from imbalanced data," *IEEE Trans. Knowl. Data Eng.*, vol. 21, no. 9, pp. 1263–1284, Sep. 2009.

[43] X.-Y. Liu, J. Wu, and Z.-H. Zhou, "Exploratory undersampling for class-imbalance learning," *IEEE Trans. Syst., Man, Cybern., B (Cybernetics)*, vol. 39, no. 2, pp. 539–550, Apr. 2009.

[44] A. Parmar, R. Katariya, and V. Patel, "A review on random forest: An ensemble classifier," in *Proc. Int. Conf. Intell. Data Commun. Technol. Internet Things*, Dec. 2018, pp. 758–763.

[45] A. S. More and D. P. Rana, "Review of random forest classification techniques to resolve data imbalance," in *Proc. 1st Int. Conf. Intell. Syst. Inf. Manage. (ICISIM)*, Oct. 2017, pp. 72–78.

[46] C. Rudin, "Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead," *Nature Mach. Intell.*, vol. 1, no. 5, pp. 206–215, May 2019.

[47] Z. C. Lipton, "The mythos of model interpretability: In machine learning, the concept of interpretability is both important and slippery," *Queue*, vol. 16, no. 3, pp. 31–57, Jun. 2018.

[48] C. Zhang, S. Bengio, M. Hardt, B. Recht, and O. Vinyals, "Understanding deep learning requires rethinking generalization," 2016, *arXiv:1611.03530*.

[49] Z. Khan, A. Gul, A. Perperoglou, M. Miftahuddin, O. Mahmoud, W. Adler, and B. Lausen, "Ensemble of optimal trees, random forest and random projection ensemble classification," *Adv. Data Anal. Classification*, vol. 14, no. 1, pp. 97–116, Mar. 2020.

[50] Z. Khan, N. Gul, N. Faiz, A. Gul, W. Adler, and B. Lausen, "Optimal trees selection for classification via Out-of-Bag assessment and sub-bagging," *IEEE Access*, vol. 9, pp. 28591–28607, 2021.

[51] M. Sebban, R. Nock, J.-H. Chauchat, and R. Rakotomalala, "Impact of learning set quality and size on decision tree performances," *IJCSS*, vol. 1, pp. 85–105, Jan. 2000.
[52] B. Efron and R. Tibshirani, *An Introduction to the Bootstrap*, vol. 57. London, U.K.: Chapman & Hall, 1993.
[53] P. Bühlmann and B. Yu, "Analyzing bagging," *Ann. Statist.*, vol. 30, no. 4, pp. 927–961, Aug. 2002.
[54] E. Fix, *Discriminatory Analysis: Nonparametric Discrimination, Consistency Properties*, vol. 1. Randolph Field, TX, USA: USAF School of Aviation Medicine, 1985.
[55] P. Cunningham and S. J. Delany, "K-nearest neighbour classifiers-a tutorial," *ACM Comput. Surveys (CSUR)*, vol. 54, no. 6, pp. 1–25, 2021.
[56] S. A. Dudani, "The distance-weighted k-Nearest-Neighbor rule," *IEEE Trans. Syst., Man, Cybern.*, vols. SMC–6, no. 4, pp. 325–327, Apr. 1976.
[57] W. S. McCulloch and W. Pitts, "A logical calculus of the ideas immanent in nervous activity," *Bull. Math. Biol.*, vol. 52, nos. 1–2, pp. 115–133, Jan. 1990.
[58] L. Breiman, "Random forests," *Mach. Learn.*, vol. 45, no. 1, pp. 5–32, 2001.
[59] V. N. Vapnik, *The nature of statistical learning theory*. New York, NY, USA: Springer-Verlag, Nov. 1997.
[60] P. S. Fader and B. G. S. Hardie, "Forecasting repeat sales at CDNOW: A case study," *Interfaces*, vol. 31, no. 3, pp. S94–S107, Jun. 2001.
[61] K. Hornik, A. Smola, A. Karatzoglou, and A. Zeileis, "Kernlab–An S4 package for kernel methods in R," *J. Stat. Softw.*, vol. 11, no. 9, pp. 1–20, Jan. 2004.
[62] A. Liaw and M. Wiener, "Classification and regression by randomforest," *R News*, vol. 2, no. 3, pp. 18–22, 2002.
[63] W. N. Venables and B. D. Ripley, *Modern Applied Statistics With S*, 4th ed., Cham, Switzerland: Springer, 2002.
[64] K. Schliep and K. Hechenbichler, "Kknn: Weighted k-Nearest neighbors," R Package Version 1.3.1, 2016. [Online]. Available: https://github.com/KlausVigo/kknn
[65] A. Gul, A. Perperoglou, Z. Khan, O. Mahmoud, M. Miftahuddin, W. Adler, and B. Lausen, "Ensemble of a subset of KNN classifiers," *Adv. Data Anal. Classification*, vol. 12, pp. 827–840, Jan. 2018.
[66] W. Adler, S. Potapov, and B. Lausen, "Classification of repeated measurements data using tree-based ensemble methods," *Comput. Statist.*, vol. 26, no. 2, pp. 355–369, Mar. 2011.
[67] T. Hothorn and B. Lausen, "Bundling classifiers by bagging trees," *Comput. Statist. Data Anal.*, vol. 49, no. 4, pp. 1068–1078, Jun. 2005.
[68] T. Hothorn and B. Lausen, "Double-bagging: Combining classifiers by bootstrap aggregation," *Pattern Recognit.*, vol. 36, no. 6, pp. 1303–1309, Jun. 2003.

ZARDAD KHAN received the master's degree (Hons.) in statistics from the University of Peshawar, Pakistan, in 2008, the M.Phil. degree in statistics from Quaid-i-Azam University, Islamabad, Pakistan, in 2011, and the Ph.D. degree in statistics from the University of Essex, U.K., in 2015. He was an Associate Professor of statistics with Abdul Wali Khan University Mardan, Pakistan. He has completed a one year postdoctoral research with the University of Essex, U.K. Currently, he is an Associate Professor of statistics with United Arab Emirates University, United Arab Emirates. His focus is on machine learning, applied statistics, computational statistics, biostatistics, and survival analysis.

AMJAD ALI received the bachelor's degree in mathematics and the master's degree in statistics from the University of Peshawar, Pakistan, in 2013 and 2017, respectively, and the M.Phil. and Ph.D. degrees from the Department of Statistics, Abdul Wali Khan University Mardan, Pakistan, in 2020 and 2023, respectively. He is currently working as a Postdoctoral Researcher at United Arab Emirates University, UAE. His research focuses on linear models, machine learning, applied statistics, causal inference, and computational statistics.

SAEED ALDAHMANI received the bachelor's degree in statistics from United Arab Emirates University, United Arab Emirates, in 2007, the master's degree in statistics from Macquarie University, Australia, in 2010, the master's degree in applied finance from Western Sydney University, Australia, in 2011, and the Ph.D. degree in statistics from the University of Essex, U.K., in 2017. He is currently an Associate Professor of statistics with United Arab Emirates University. His focus is on graphical models, applied statistics in finance, and computational statistics.

HENRIK NORDMARK is the Director of Science, Data, and Innovation with Profusion Media Ltd., an international data science consultancy. He is also a Visiting Fellow in mathematical sciences with the University of Essex. With extensive experience in data science, analytics, artificial intelligence, and innovation, he has contributed significantly to the field. He has authored on topics, such as algorithmic fairness, emphasizing the importance of unbiased algorithms in technology design.

BERTHOLD LAUSEN is currently a Professor of data science with the School of Mathematics, Statistics and Actuarial Science (SMSAS), University of Essex, Colchester, U.K. Since 2014, he has been leading on the introduction and development of data science education with the University of Essex. He led on the introduction of various graduate and postgraduate programs in data science, funded by the Department for Digital, Culture, Media and Sport (DCMS) and the Office for Artificial Intelligence (OAI), U.K., providing funding for providers to develop postgraduate conversion courses in artificial intelligence and data science. Since 2015, he has been contributing and named as a Researcher and a Co-Investigator to the ESRC Business and Local Government Data Research Centre (BLG DRC), University of Essex. Since 2014, he has been the Principal Investigator to Knowledge Transfer Partnerships (KTP) funded by Innovate U.K., with the industrial partners Profusion, London, and Mondaq, the Innovation Centre, and the University of Essex. He has published more than 100 papers in international journals, conference proceedings, and edited books. His research interests include artificial intelligence, biostatistics, classification, clinical research, data science, machine learning, and mathematics education. He was the President (2018–2019), the President-Elect (2016–2017), and the Past President (2020–2021) of the International Federation of Classification Societies (IFCS); the former President of the Data Science Society (GfKl), from 2013 to 2019; and the former Vice President (1995–2001 and 2004–2013) and the founding Vice President (2015–2018) of European Association for Data Science (EuADS). Since 2019, he has been a Board Member of EuADS. From 2013 to 2019, he chaired the Stakeholder Group of the European Conference of Data Analysis (ECDA) and the Chair of the Scientific Program Committee of the first ECDA Conference 2013, Luxembourg.